

VIBRATION TRANSMISSION THROUGH ROLLING ELEMENT BEARINGS. PART III: GEARED ROTOR SYSTEM STUDIES

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This paper extends the proposed bearing matrix formulation of Parts I and II to analyze the overall dynamics of a geared rotor system which includes a spur gear pair, shafts, rolling element bearings, a prime mover and a load (attached to the geared rotor system through flexible torsional couplings), a rigid or flexible casing, and compliant or massive mounts. Linear time-invariant, discrete dynamic models of a generic geared rotor system with proportional viscous damping are developed, by using lumped parameter and dynamic finite element techniques, which are then used to predict the vibration transmissibility through bearings and mounts, casing vibration motion, and dynamic response of the internal rotating system. Each rotating shaft is modeled as an Euler beam in the lumped parameter model and as a Timoshenko beam in the dynamic finite element model, but the gyroscopic moment is not included. Eigensolution and forced harmonic response studies due to rotating mass unbalance or kinematic transmission error excitation for the following example cases are obtained by using the formulation, and the results are compared with those of simple models currently available in the literature and/or experiment: case I, a single-stage rotor system with flexibly mounted rigid casing consisting of two bearings as a special case of the geared rotor system; case II, a spur gear pair drive supported by four bearings installed in a flexibly mounted rigid casing; and case III, an experimental set-up consisting of a high-precision gear and pinion, and four identical rolling element bearings contained in a flexible casing mounted rigidly on a massive foundation. Analytical predictions show that the theory is indeed capable of predicting bearing and mount moment transmissibilities in addition to the force transmissibilities. Also, flexural vibrations of the casing plate are predicted well as the theory is in good agreement with measurements made on case III; such predictions are not provided by simple models due to inadequate bearing formulation.

1. INTRODUCTION

The focus of this paper is on the dynamic analysis of the geared rotor system shown in Figure 1, which includes a spur gear pair, shafts, rolling element bearings, a motor, a load, a casing and flexible or rigid mounts. For this purpose, discrete vibration models are developed and used to predict vibration transmission through the bearings. Also, the effects of the bearing, casing and mount dynamics on the internal rotating system dynamic characteristics are investigated. Traditional analysis approaches [1–5] in the gear dynamics area have been concentrated on the internal rotating system and have excluded dynamic effects of the casing and flexible mounts. Moreover, simple bearing models have been typically used in which either ideal boundary conditions on the shaft or translational spring

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elements have been assumed. Some of the limitations associated with current bearing models have been discussed thoroughly in Parts I and II of this paper [6, 7]. For instance, simple bearing formulations cannot explain how the vibratory motion may be transmitted from the rotating shafts to casing and support structures and, moreover, cannot adequately predict the effects of bearing, casing and mount dynamics on the internal rotating system. In Part I of this paper a new mathematical model for the precision rolling element bearings was presented to clarify this issue qualitatively and quantitatively, and in Part II the proposed bearing stiffness matrix $[K]_{bm}$ formulation was used to analyze the vibration transmission problem in a generic shaft-bearing-casing-mount component and stability of the proposed bearing system model. In Part II also a comprehensive analysis was provided of the mechanics of vibration transmission through the bearing due to purely shaft bending motion, with emphasis on the bearing as a component. This bearing formulation has already been validated through several example cases.

In this paper the proposed bearing formulation of Parts I and II is extended to examine vibration transmissibility in rotating mechanical equipment through two generic systems, which are a spur gear pair drive and a single-stage rotor system with rolling element

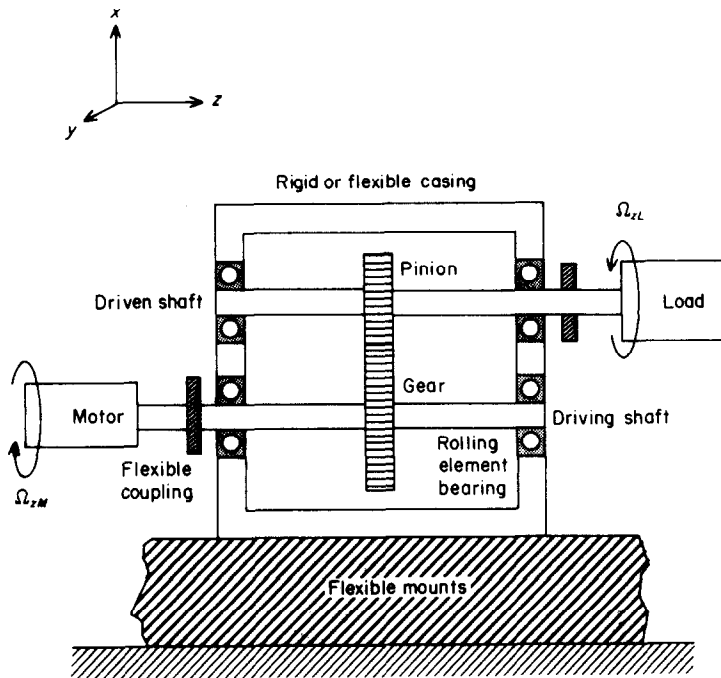


Figure 1. Schematic of a geared rotor system with flexible shafts and rigid spur gear and pinion, and supported by four rolling element bearings on a flexibly mounted casing. The shaft at the motor end is driven at mean speed $\Omega_{z,M}$, which in turn drives the load at mean speed $\Omega_{z,L}$. This physical system is symmetric about a plane intersecting both the driving and driven shafts. The y axis is assumed to be parallel to the gear mesh line of action in the pressure angle direction at mesh point. This schematic is used for example cases II and III.

bearings, as shown in Figures 1 and 2 respectively. It should be noted that Figure 2 can be treated as a special case of Figure 1—this will be illustrated later. The physical systems are assumed to be driven at mean speed $\Omega_z = \Omega_{z,M}$ in Figure 1 and Ω_z in Figure 2; subscript M refers to the motor end. Each shaft is supported by two identical rolling element bearings which are modeled as stiffness matrices $[K]_{bm}$ of dimension six, as proposed in Part I of this paper. These generic systems may be excited by motor and/or torque load fluctuations, rotating mass unbalances or gear kinematic transmission error $e(t)$ [1, 8]; here $e(t)$ is

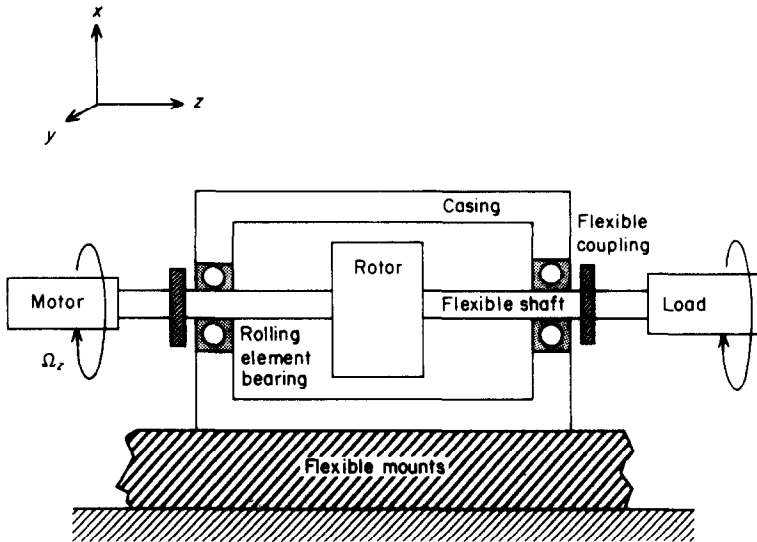


Figure 2. Schematic of the single-stage rotor system with flexible shaft rotating at mean speed Ω_r and supported by two identical rolling element bearings on a flexibly mounted rigid casing. This physical system is symmetric about a plane intersecting the shaft axial axis and parallel to the x - z plane. This schematic is used for example case I.

defined as the deviation of the relative gear-pinion angular position from its relative ideal location. Typical excitation at the shaft is defined by an alternating load vector $\{f(t)\}_{sa} = \{F_{jsa}(t), T_{jsa}(t)\}^T = \{f(t)\}_s - \{f\}_{sm}$; $j = x, y, z$, where $F_{jsa}(t)$ and $T_{jsa}(t)$ are the alternating force and torque respectively, $\{f(t)\}_s$ is the total load vector of dimension six, $\{f\}_{sm}$ represents the mean load vector, and superscript T implies the transpose. In the case of the geared drive, $\{f(t)\}_{sa}$ consists of $e(t)$ at the mesh point. Additionally, in the case of the single-stage rotor system, transverse forces $F_{xsa}(t)$ and $F_{ysa}(t)$ due to mass unbalances are of interest. Note that in the dynamic analysis $\{f\}_{sm}$ and bearing preloads are not included, as they do not appear in the governing equations of the linear vibration model but are used for computing $[K]_{bm}$. Other effects such as bearing coupling coefficients, motor and load inertia, and casing and mount dynamics are considered, as the emphasis is on the overall system behavior. (A list of symbols is given in the Appendix.)

2. REVIEW OF THE LITERATURE

The available literature on bearing models commonly used for internal rotor and gear dynamic analyses has already been discussed in Parts I and II of this paper [6, 7]. Although there are numerous publications [1-5, 9-12] on the modal analysis of geared drives and single or multiple-stage rotor systems, very little has been reported directly on the force transmissibility through bearings, and the dynamic effects of bearing, casing and mounts on the internal rotating system.

2.1. CASING AND MOUNT DYNAMICS

A comprehensive review of the available vibration studies of casing and mounts, without the internal rotating system, has been given by Lim and Singh in 1989 [13]. Other related publications not included in reference [13] are by Gaul and Mahrenholtz [14] in 1984, and Smith [15] in 1988, who have developed lumped parameter models of flexible machine

foundations, excluding the internal rotating system dynamics. Their studies report rigid-body translational and rotational vibration responses of the casing when excited by forces and/or moments. However, these studies are not specifically directed towards geared rotor systems. In addition, Lim *et al.* [16] in 1989 have also performed a dynamic finite element analysis of a helicopter gearbox, excluding the internal rotating system. Results again show that casing rigid-body vibration modes are dominant over the lower frequencies for a flexibly mounted casing, but numerous casing plate elastic modes are observed at higher frequencies.

2.2. GEAR AND ROTOR DYNAMICS

Current gear and rotor dynamic models include only the internal rotating system and simple bearing models, and typically exclude casing and mount flexibilities [1–5, 9–12]. In a few instances casing and mounts have also been included, but restricted to cases in which only purely radial and/or axial force on the bearing and translational rigid-body motion on the casing are modeled, as summarized in reference [13] for geared systems and as given in references [20–22] for rotor dynamic models. Such models are still inadequate in explaining how the vibration is transmitted from the internal vibration sources to the casing and into the machine foundation, as witnessed in previous experiments [17–19]. Moreover, the overall dynamic behavior of these systems have been investigated only partially, due to the incomplete description of $[K]_{bm}$.

3. ASSUMPTIONS AND OBJECTIVES

Linear discrete vibration models of the geared and single-stage rotor systems shown in Figures 1 and 2 are used, incorporating an appropriate $[K]_{bm}$ in a manner similar to that of Part II [7]. The linear model assumption is valid under the high mean torque load and bearing preloads [23] which are being considered here. The stiffness coefficients of $[K]_{bm}$ are evaluated by using the analytical expressions presented in Part I [6]. Each rotating shaft is modeled as an Euler beam in the lumped parameter model and as a Timoshenko beam in the dynamic finite element model, but the effect of the gyroscopic moment on the shaft dynamics is not included. The casing is assumed to be rigid in the lumped parameter model and flexible in the dynamic finite element model. The assumption of a rigid casing is valid for many practical designs, as it may be massive and the rotational speeds may be sufficiently low to avoid significant elastic deformation. The governing equations for both discrete vibration models can be given in the general matrix form as

$$[M]\{\ddot{q}(t)\}_a + [C]\{\dot{q}(t)\}_a + [K]\{q(t)\}_a = \{f(t)\}_a, \quad (1)$$

where $[M]$, $[C]$ and $[K]$ are the system mass, damping and stiffness matrices respectively, and $\{q(t)\}_a$ and $\{f(t)\}_a$ are defined as the generalized alternating displacement and applied load vectors respectively. As in Part II of this paper, $\{f\}_{bm}$ and bearing preloads are also assumed constant to ensure a time-invariant $[K]_{bm}$ matrix. Energy equivalent viscous damping matrix $[C]_b = \sigma[K]_{bm}$ is assumed for the energy dissipation mechanism in the bearing, where σ is the Rayleigh damping proportionally constant. Other features of the proposed theory given in Part II of this paper are retained.

The specific objectives of this study are as follows: (i) to incorporate the proposed bearing matrix $[K]_{bm}$, developed in Part I, in discrete vibration models of the generic geared system of Figure 1 as described by equation (1), by using lumped parameter and/or dynamic finite element methods; (ii) to conduct overall system studies by calculating eigensolutions and forced harmonic responses; (iii) to evaluate the effects of casing and mount dynamics on the internal rotating system, and predict vibration transmission

through bearings; (iv) to examine the following three example cases: I, a single-stage rotor system with flexible shaft supported by two identical rolling element bearings on rigid casing and flexible mounts as shown in Figure 2; II, a spur gear pair system with motor and load inertias attached to two flexible shafts supported by four rolling element bearings on flexibly and rigidly mounted rigid casing as shown in Figure 1; III, case II with flexible casing and rigid mounts; (v) to demonstrate the advantages of the formulation over the existing vibration models.

4. THEORY

4.1. METHOD A: LUMPED PARAMETER MODEL

4.1.1. Equations of motion

Both physical systems, as shown in Figures 1 and 2, are discretized by using the lumped parameter technique to yield equation (1) through the Lagrange's equations of motion [24]

$$\frac{d}{dt} \left(\frac{\partial E_T}{\partial \dot{q}_w} - \frac{\partial E_U}{\partial \dot{q}_w} \right) - \frac{\partial E_T}{\partial q_w} + \frac{\partial E_U}{\partial q_w} + \frac{\partial E_C}{\partial \dot{q}_w} = \tilde{F}_w, \quad w = 1, 2, 3, \dots \quad (2)$$

Here E_T and E_U are the kinetic and potential energies respectively, $E_C = \frac{1}{2} \{\dot{q}\}^T [C] \{\dot{q}\}$ is the Rayleigh's dissipation function and $\tilde{F}_w$ is the generalized force. The total system potential energy E_U and kinetic energy E_T are obtained by adding the energies of each of the system components, which are derived in subsequent sections. When applied to Figure 1, equation (2) will lead to equation (1). It is then rearranged in terms of the following $\{q\}$ to obtain the partitioned system matrices:

$$\{q(t)\}_a = \{ \{u(t)\}_{sa}^T \quad \{u(t)\}_{ca}^T \mid \{\theta(t)\}_{Ra}^T \quad \{\theta(t)\}_{ca}^T \mid \{\theta(t)\}_{sa}^T \}^T, \quad (3a)$$

$$[M] = \text{diag} \{ \{m\}_s^T \quad \{m\}_c^T \mid \{I\}_R^T \quad \{I\}_c^T \mid \{I\}_s^T \}, \quad (3b)$$

$$[K] = \begin{bmatrix} [K]_{11} & [K]_{12} & [K]_{13} \\ [K]_{21} & [K]_{22} & [K]_{23} \\ [K]_{31} & [K]_{32} & [K]_{33} \end{bmatrix}, \quad [C] = \sigma [K]. \quad (3c)$$

Here $\{\theta(t)\}_{Ra}$ (subscript R indicates rotor) consists of the alternating angular displacement of all rotors including gears, motor and load; $\{m\}^T$ and $\{I\}^T$ are the lumped mass and inertia row vectors respectively; and the operator $\text{diag} \{ \}$ transforms the row vector into a diagonal matrix, with components of the vector corresponding to the diagonal elements. The stiffness submatrices $[K]_{wj} = [K]_{jw}^T$, $w, j = 1, 2, 3$, consist of the appropriate terms corresponding to the partitioned $\{q(t)\}_a$. For example, $[K]_{11}$ couples $\{ \{u(t)\}_{sa}^T \quad \{u(t)\}_{ca}^T \}^T$ degrees of freedom to itself, and $[K]_{12}$ provides a coupling between $\{ \{u(t)\}_{sa}^T \quad \{u(t)\}_{ca}^T \}^T$ and $\{ \{\theta(t)\}_{Ra}^T \quad \{\theta(t)\}_{ca}^T \}^T$ degrees of freedom.

Now assume that the rotary inertia of each shaft lumped mass is negligible. The system matrices given in equation (3) are therefore simplified by eliminating $\{\theta(t)\}_{sa}$ degrees of freedom from $\{q(t)\}_a$ in equation (3a). The resulting mass and stiffness matrices are

$$[M] = \text{diag} \{ \{m\}_s^T \quad \{m\}_c^T \mid \{I\}_R^T \quad \{I\}_c^T \}, \quad (4a)$$

$$[K] = \begin{bmatrix} [K]_{11} - [K]_{13}[K]_{33}^{-1}[K]_{31} & [K]_{12} - [K]_{13}[K]_{33}^{-1}[K]_{32} \\ [K]_{21} - [K]_{23}[K]_{33}^{-1}[K]_{31} & [K]_{22} - [K]_{23}[K]_{33}^{-1}[K]_{32} \end{bmatrix}. \quad (4b)$$

This analytical lumped parameter formulation will be used for example cases I and II. It may be noted that if the effect of the rotary inertias is required equation (3) can be used instead.

4.1.2. System kinetic energy

A flexible shaft of length L_s with rigid rotor of mass m_R is subdivided into n_s number of segments of equal length $L_e = L_s/n_s$ with lumped masses at both ends of each segment, as shown in Figure 3(a) for a single-spur gear mesh geared drive. Each lumped mass has three translational and three rotational degrees of freedom, as illustrated in Figure 3(b). The total system kinetic energy E_T is given by $\frac{1}{2} \{\dot{q}\}^T [M] \{\dot{q}\}$. The mass matrix $[M]$ is diagonal and consists of the lumped masses and inertias.

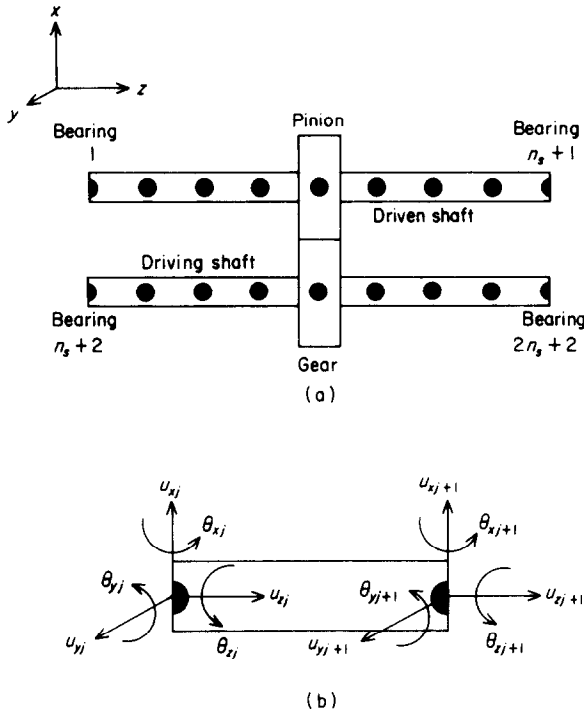


Figure 3. Lumped parameter model of the flexible shafts with rigid rotors. Each shaft is supported by two identical bearings on both ends. (a) Internal rotating system of a typical geared rotor system of Figure 1; (b) discretization of a shaft segment and the degrees of freedom associated with each lumped mass.

4.1.3. Shaft stiffness matrix

By using the direct stiffness approach, the stiffness matrix $[K]_s^e$ of dimension 12 corresponding to the alternating displacement vector $\{q(t)\}_{sa}^e = \{u_{xj}(t), u_{xj+1}(t), u_{yj}(t), u_{yj+1}(t), u_{zj}(t), u_{zj+1}(t), \theta_{yj}(t), \theta_{yj+1}(t), \theta_{xj}(t), \theta_{xj+1}(t), \theta_{zj}(t), \theta_{zj+1}(t)\}^T$ of a generic shaft segment is found to be given by

$$[K]_s^e = \begin{bmatrix} [K_{uu}]_s^e & [K_{u\theta}]_s^e \\ [K_{\theta u}]_s^e & [K_{\theta\theta}]_s^e \end{bmatrix}, \quad (5)$$

the non-zero elements of $[K_{uu}]_s^e$, $[K_{u\theta}]_s^e = [K_{\theta u}]_s^{eT}$ and $[K_{\theta\theta}]_s^e$ of dimension six being given by the following equations (6), (7) and (8) respectively:

$$k_{11} = k_{22} = k_{33} = k_{44} = -k_{12} = -k_{21} = -k_{34} = -k_{43} = 12EI/L_e^3,$$

$$k_{55} = k_{66} = -k_{56} = -k_{65} = AE/L_e, \quad (6)$$

$$k_{11} = -k_{22} = k_{33} = -k_{44} = k_{12} = -k_{21} = k_{34} = -k_{43} = 6EI/L_e^2, \quad (7)$$

$$k_{11} = k_{22} = k_{33} = k_{44} = 4EI/L_e, \quad k_{12} = k_{21} = k_{34} = k_{43} = 2EI/L_e,$$

$$k_{55} = k_{66} = -k_{56} = -k_{65} = GJ/L_e. \quad (8)$$

Here E is the modulus of elasticity, G is the shear modulus of elasticity, I is the moment of inertia, and J is the polar moment of inertia of the shaft. The lumped stiffness matrix $[K]_s$ corresponding to $\{q(t)\}_{sa} = \{u_{xj}(t), u_{yj}(t), u_{zj}(t), \theta_{yj}(t), \theta_{xj}(t), \theta_{zj}(t)\}^T$, $j = 1, 2, 3, \dots$, for the shaft is constructed by the superposition of all $[K]_s^e$ matrices and merging terms associated with each degree of freedom:

$$[K]_s = \begin{bmatrix} [K_{uu}]_s & [K_{u\theta}]_s \\ [K_{\theta u}]_s & [K_{\theta\theta}]_s \end{bmatrix}. \quad (9)$$

The non-zero elements of $[K_{uu}]_s$, $[K_{u\theta}]_s = [K_{\theta u}]_s^T$ and $[K_{\theta\theta}]_s$ of dimension $3(n_s + 1)$ are similarly given by the following equations (10), (11) and (12) respectively:

$$k_{11} = k_{22} = k_{3n_s+1, 3n_s+1} = k_{3n_s+2, 3n_s+2} = 12EI/L_e^3, \quad k_{33} = k_{3n_s+1, 3n_s+1} = AE/L_e, \quad (10a)$$

$$k_{w_I+3, w_I} = k_{w_I, w_I+3} = -12EI/L_e^3, \quad w_I = 1, 2, 4, 5, 7, 8, \dots, 3n_s-1, \quad (10b)$$

$$k_{w_{II}, w_{II}} = 24EI/L_e^3, \quad w_{II} = 4, 5, 7, 8, 10, 11, \dots, 3n_s-1, \quad (10c)$$

$$k_{w_{III}+3, w_{III}} = k_{w_{III}, w_{III}+3} = -AE/L_e, \quad w_{III} = 3, 6, 9, \dots, 3n_s, \quad (10d)$$

$$k_{w_{IV}, w_{IV}} = 2AE/L_e, \quad w_{IV} = 6, 8, 12, \dots, 3n_s, \quad (10e)$$

$$k_{11} = k_{22} = -k_{3n_s+1, 3n_s+1} = -k_{3n_s+2, 3n_s+2} = 6EI/L_e^2, \quad (11a)$$

$$-k_{w_I+3, w_I} = k_{w_I, w_I+3} = 6EI/L_e^2, \quad w_I = 1, 2, 4, 5, 7, 8, \dots, 3n_s-1, \quad (11b)$$

$$k_{11} = k_{22} = k_{3n_s+1, 3n_s+1} = k_{3n_s+2, 3n_s+2} = 4EI/L_e, \quad k_{33} = k_{3n_s+1, 3n_s+1} = GJ/L_e, \quad (12a)$$

$$k_{w_I+3, w_I} = k_{w_I, w_I+3} = 2EI/L_e, \quad w_I = 1, 2, 4, 5, 7, 8, \dots, 3n_s-1, \quad (12b)$$

$$k_{w_{II}, w_{II}} = 8EI/L_e, \quad w_{II} = 4, 5, 7, 8, 10, 11, \dots, 3n_s-1, \quad (12c)$$

$$k_{w_{III}+3, w_{III}} = k_{w_{III}, w_{III}+3} = -GJ/L_e, \quad w_{III} = 3, 6, 9, \dots, 3n_s, \quad (12d)$$

$$k_{w_{IV}, w_{IV}} = 2GJ/L_e, \quad w_{IV} = 6, 8, 12, \dots, 3n_s. \quad (12e)$$

Note that here $[K]_s$ is only for a shaft, of dimension $6(n_s + 1)$. The same formulation should be applicable to a geared system with multiple shafts, where each shaft stiffness matrix is still given by $[K]_s$. Accordingly, the shaft potential energy E_{Us} is given as

$$E_{Us} = \frac{1}{2} \{q\}_s^T [K]_s \{q\}_s. \quad (13)$$

Formulations for the motor and load rotary inertias, and flexible coupling torsional stiffness will be given in the later sections on example cases.

4.1.4. Gear mesh stiffness matrix

For the generic geared rotor system with a spur gear pair, the driving and driven shafts are coupled via a linear, time-invariant spur gear mesh stiffness $k_h \neq k_h(t)$ for a high drive torque. One can now define the gear mesh coupling stiffness matrix $[K]_h$ as follows, where coupled torsional and translational motions of the gear and pinion are defined by $\{u_{yGa}(t), u_{yPa}(t), \theta_{zGa}(t), \theta_{zPa}(t)\}^T$:

$$[K]_h = \begin{bmatrix} k_h & -k_h & \frac{k_h d_G}{2} & -\frac{k_h d_P}{2} \\ -k_h & k_h & -\frac{k_h d_G}{2} & \frac{k_h d_P}{2} \\ \frac{k_h d_G}{2} & -\frac{k_h d_G}{2} & \frac{k_h d_G^2}{4} & -\frac{k_h d_P d_G}{4} \\ -\frac{k_h d_P}{2} & \frac{k_h d_P}{2} & -\frac{k_h d_P d_G}{4} & \frac{k_h d_P^2}{4} \end{bmatrix}. \quad (14)$$

Here d_G and d_P are the gear and pinion diameters respectively; subscripts G and P refer to gear and pinion, respectively. The y axis is taken to be parallel to the gear mesh force which is along the pressure angle direction at the mesh point. The potential energy E_{Uh} in this case is

$$E_{Uh} = \frac{1}{2} \{u_{yGa}, u_{yPa}, \theta_{zGa}, \theta_{zPa}\} [K]_h \begin{Bmatrix} u_{yGa} \\ u_{yPa} \\ \theta_{zGa} \\ \theta_{zPa} \end{Bmatrix}. \quad (15)$$

4.1.5. Flexible mount stiffness matrix

The flexible mounts are represented by a diagonal stiffness matrix $[K]_v$ corresponding to $\{q(t)\}_{ca}$, which consists of effective stiffness coefficients k_{vw} , $w = x, y, z, \theta_x, \theta_y, \theta_z$. In this modeling procedure it is assumed that there is no coupling between the casing rigid-body degrees of freedom due to the flexibility of the mounts. Accordingly, one has

$$E_{Uv} = \frac{1}{2} \{q\}_{ca}^T [K]_v \{q\}_{ca}. \quad (16)$$

4.1.6. Bearing stiffness matrix

The proposed bearing stiffness matrix $[K]_{bm}$ corresponding to the bearing degrees of freedom $\{\delta_{wa}(t), \beta_{wa}(t)\}^T$, $w = x, y, z$, has been formulated in Part I [6]. The corresponding potential energy E_{Ub} is

$$E_{Ub} = \frac{1}{2} \{\delta_{waj}, \beta_{waj}\} [K]_{bm} \begin{Bmatrix} \delta_{waj} \\ \beta_{waj} \end{Bmatrix}. \quad (17)$$

Here, $\{\delta_{waj}(t), \beta_{waj}(t)\}^T$ may be expressed in terms of $\{q(t)\}_{saj}$ and $\{q(t)\}_{ca}$ through a coordinate transformation for the j th bearing located at $\mathbf{R}_j = \{x_j, y_j, z_j\}$ from the casing center of mass. For the geared rotor system, four bearings are denoted by $j = 1, n_s + 1, n_s + 2$ and $2n_s + 2$, as shown in Figure 3(a), while for the single-stage rotor system two bearings are

denoted by $j=1$ and n_s+1 only. Under the assumed small angle approximation, $\{\delta_{waj}(t), \beta_{waj}(t)\}^T$ for the j th bearing is

$$\begin{Bmatrix} \delta_{waj} \\ \beta_{waj} \end{Bmatrix} = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & z_j & -y_j \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & -z_j & 0 & x_j \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & y_j & -x_j & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} \{q\}_{saj} \\ \{q\}_{ca} \end{Bmatrix}. \quad (18)$$

4.2. METHOD B: DYNAMIC FINITE ELEMENT FORMULATION

The dynamic finite element technique of incorporating $[K]_{bm}$ in equation (1) is employed especially when the casing plate is elastically deformable over the frequency range of interest, as discussed previously in Part II [7]. In this method, $[K]_{bm}$ is implemented in the dynamic finite element model as a generalized stiffness matrix. This stiffness element provides the only coupling between the flexible shaft and casing at the appropriate nodal points on both components corresponding to the bearing locations. Additionally, the gear mesh stiffness k_h in Figure 1 behaves like a linear translational spring. Since the coupling between torsional and transverse shaft vibrations is due to k_h only, the corresponding portion of the stiffness matrix is also given by equation (14). Our finite element formulation uses conventional structural elements typically available in commercial software programs [25] for the shaft, casing plate and mounts—this will be illustrated for example cases I, II and III. Other features of this method are similar to those discussed earlier in section 4.1.

4.3. OTHER METHODS

Alternative techniques of formulating the geared rotor system dynamic problems such as finite difference (which is similar to method B), flexibility, component mode synthesis and transfer matrix formulations are also possible. It may be noted that Berman [26] has philosophically proposed application of component mode synthesis to geared problems. Other researchers [27, 28] have utilized the transfer matrix method in rotor dynamic problems. However, such models need to be modified or extended in order to solve the overall dynamic problems discussed in this paper. Direct applications of these alternative methods [26–30] are beyond the scope of this paper and are left for further research.

5. EXAMPLE CASE I: SINGLE-STAGE ROTOR SYSTEM WITH RIGID CASING AND FLEXIBLE MOUNTS

Consider the single-stage rotor system shown in Figure 2, where two highly preloaded identical deep groove ball bearings support a flexible rotating shaft (see Table 1 for bearing parameters). The shaft is assumed to be subjected to a mean drive torque $T_{zsm} \neq T_{zsm}(t)$ and rotating rotor mass unbalance forces

$$\{F_{xsa}(t) = F_{xsa1}(\Omega_z) e^{i\Omega_z t}, F_{ysa}(t) = F_{ysa1}(\Omega_z) e^{i(\Omega_z t - \pi/2)}\}^T.$$

It is also assumed to be transversely decoupled from the motor and load due to the flexible torsional couplings.

A lumped parameter model with n_s+2 lumped masses is developed according to the theory given in section 4.1. The bearing stiffness matrix for an axially preloaded ball bearing has significant stiffness coefficients k_{bxx} , k_{byy} , k_{bzz} , $k_{b\theta_x\theta_x}$, $k_{b\theta_y\theta_y}$, $k_{b\theta_z\theta_z}$ and $k_{b\theta_x\theta_y}$. The system equation (1) may be simplified to a two-dimensional vibratory system with

TABLE 1

Design parameters for deep groove ball bearings used for example cases†

Load-deflection exponent n	3/2
Load-deflection constant, $K_n(\text{N/m}^n)$	7.5E9
Number of rolling elements, Z	12
Radial clearance, r_L (mm)	0.005–0.02
Pitch diameter (mm)	38.5
A_0 (mm)‡	0.0625
Unloaded contact angle, α_0	23°–47°

† These bearings are currently being used in a NASA gear test facility [32]; also see examples cases II and III.

‡ Unloaded distance between inner and outer raceway groove curvature centers.

$\text{DOF} = n_s + 4$ and $\{q(t)\}_a = \{u_{xsa}(t), u_{xca}(t), \theta_{yRa}(t), \theta_{yca}(t)\}^T$ due to the absence of external force $F_{za}(t)$ on the system, and due to system symmetry. The system matrices and other details were given in a recent conference paper by the authors [31]. The steady state solution will always result in the synchronous whirling motion of the internal rotating shaft. Our proposed vibration model can also be readily reduced to conventional rotor dynamic models [9–12], excluding the gyroscopic moment, internal structural damping, dissimilar bearings and other secondary effects, by retaining only k_{bxx} , k_{byy} and k_{bzz} in our bearing formulation.

The natural frequencies ω_j and modes ϕ_j of Figure 2 with system parameters given in Table 2 are obtained numerically, and are compared with two simple lumped parameter vibration models and a dynamic finite element model (FEM) in Table 3. Our proposed theory and FEM are in excellent agreement with each other as the deviations are within

TABLE 2

System parameters of example case I: single-stage rotor system shown in Figure 2

Rotor mass, m_R (kg) and inertia, I_{yR} (kg m ²)	5.0, 1.24E–2
Shaft mass, m_s (kg) and length, L_s (m)	1.03, 4.2E–1
Casing mass, m_c (kg) and inertia, I_{yc} (kg m ²)	148, 4.27
Shaft flexural rigidity, EI (N m ²)	1.626E3
Bearing axial preload, F_{zbm} (N)	238
$k_{bxx}, k_{bxy}, k_{byy}, k_{b\theta_y}, k_{b\theta_x}$ †	1.37E8, –1.96E6, 4.27E4
$k_{vxx}, k_{v\theta_y}$	4.0E8, 1.8E7

† Computed k_{bwy} as proposed in Part I; only relevant bearing coefficients are listed.

TABLE 3

Comparison of natural frequencies in Hz for example case I

Mode	Proposed theory	FEM	Δ_1 (%)	Simple models		
				Without casing	With casing	Δ_2 (%)
1	97.2	97.2	0.0	69.5	69.4	28.6
2	262	262	0.0	Not predicted	262	0.0
3	309	309	0.0	297	294	4.9
4	332	334	0.6	Not predicted	329	1.5
5	1405	1365	–2.9	1442	1443	–5.7
6	1755	1690	–3.8	1452	1453	14.0

Δ_1 (%) = $100 \times (\text{FEM} - \text{proposed theory}) / \text{FEM}$; Δ_2 (%) = $100 \times (\text{FEM} - \text{simple model}) / \text{FEM}$.

$\pm 4\%$ for the first six natural frequencies. The simple model with casing and mount dynamics included predicts lower natural frequencies as compared to our prediction, except for $\hat{\omega}_5$; here the symbol $\hat{\omega}$ implies estimation by using a simple formulation. Furthermore, the simple model without casing and mounts cannot predict all the modes below 2 kHz. The forced harmonic response due to the rotating rotor mass unbalance force, and the force $R_{F_{xba}, F_{xsa}}(\Omega_z)$ and moment $R_{M_{yba}, F_{xsa}}(\Omega_z)$ transmissibility spectra are also computed and compared with those of the simple formulation in reference [31].

6. EXAMPLE CASE II: GEARED ROTOR SYSTEM WITH RIGID CASING AND FLEXIBLE MOUNTS

6.1. BEARING ANALYSIS

Now consider the geared rotor system in Figure 1 with a flexibly mounted rigid casing, the bearing and other system parameters of which are given in Tables 1 and 4 respectively.

TABLE 4

System parameters of example case II: geared rotor system shown in Figure 1†

Gear and pinion mass, $m_P = m_G$ (kg)	0.5
Gear and pinion rotary inertias, $I_{yG} = I_{yP}$, $I_{zG} = I_{zP}$ (kg m ²)	1.5E-4, 3.0E-4
Number of gear/pinion teeth	28
Gear/pinion pressure angle (degrees)	20°
Shaft mass, m_s (kg) and length, L_s (m)	2.8, 2.54E-1
Casing mass, m_c (kg) rotary inertias, I_{xc} , I_{zc} (kg m ²)	77, 1.5, 1.9
Shaft flexural rigidity, EI (N m ²)	1.25E4
Effective torsional stiffness, k_{TM} , k_{TL} (Nm/rad)	6.05E3, 2.45E4
Motor and load inertia, I_{zM} , I_{zL}	1.00E-1, 3.35
Gear mesh stiffness, k_h (N/m)	1.0E8
Mean axial bearing displacement, δ_{zm} (N)	6.00E-4
Mean input torque, T_{zbm} (Nm)	72
k_{byy} , $k_{b\theta_x}$, $k_{b\theta_y}$, ‡	9.7E8, 6.0E5, 1.0E6
k_{vy} , $k_{v\theta_x}$, $k_{v\theta_y}$	9.0E8, 1.4E7, 2.4E7

† The internal rotating spur gear pair is currently being used in a NASA gear test facility [32].

‡ Only relevant bearing stiffness coefficients are tabulated.

Four ball bearings which support the two shafts are subjected to identical mean axial displacement δ_{zm} . The spur gear pair drive is driven by a mean torque $T_{zsm} \neq T_{zsm}(t)$ which also generates mean radial bearing force F_{ybmj} and moment M_{xbmj} . The stiffness matrix $[K]_{bm}$ for each bearing under these loads has significant coefficients k_{bxx} , k_{byy} , k_{bzz} , $k_{b\theta_x}$, $k_{b\theta_y}$, $k_{b\theta_z}$, $k_{b\theta_x}$, k_{byz} and $k_{bz\theta_x}$. A set of governing non-linear algebraic equations, consisting of three bearing load-displacement relations for each bearing as given in Part I [6] and from the shaft bending theory is given by

$$\begin{aligned} M_{xbmw_{II}} &= T_{zsm}L_s/(8d_G) - 2EI(2\beta_{xmw_{II}} + \beta_{xmw_{II}} + 3(\delta_{ymw_{II}} - \delta_{ymw_{II}})/L_s)/L_s, \\ M_{xbmw_{II}} &= -M_{xbmw_{II}}, \quad \text{for } w_{II} = n_s + 1 \quad \text{if } w_I = 1, \quad \text{and} \quad w_{II} = 2n_s + 2 \quad \text{if } w_I = n_s + 2. \end{aligned} \quad (19)$$

Due to the physical symmetry, one can assume that $\delta_{ymw_I} = \delta_{ymw_{II}}$ and $\beta_{xmw_I} = -\beta_{xmw_{II}}$, which simplifies the non-linear algebraic equation set for each bearing to

$$M_{xbm} = T_{zsm}L_s/(8d_G) - 2EI\beta_{xm}/L_s, \quad F_{ybm} = T_{zsm}/d_G, \quad (20)$$

where M_{xbm} and F_{bym} are expressed in terms of a constant δ_{zm} , and variables δ_{ym} and β_{xm} . Solutions of these two equations may then be used to compute $k_{bij}(\delta_{zm}, \delta_{ym}, \beta_{xm})$ directly.

6.2. VIBRATION MODELS

A lumped parameter model of Figure 1 is developed in accordance with section 4.1. Each shaft is divided into n_s segments, as shown in Figure 3(a). The system matrices of equation (4) are simplified by neglecting the longitudinal shaft motion and casing rigid-body degrees of freedom $u_{zca}(t)$, $u_{xca}(t)$ and $\theta_{yca}(t)$. Both driving and driven shafts are assumed to be coupled to the motor and load respectively through flexible torsional couplings. Therefore, only the motor I_{zM} and load I_{zL} rotary inertias are considered, and the exterior portion of the shafts beyond the flexible couplings are modeled as purely torsional stiffness elements which are then combined with the flexible coupling stiffnesses. The system is excited by the static transmission error $e(t)$ at the mesh point which generates gear and pinion force $F_h(t) = k_h e(t)$, parallel to the line of action, and torque $T_{hG}(t) = (1/2)d_G k_h e(t)$ or $T_{hP}(t) = (1/2)d_P k_h e(t)$, about the axial z -axis. The mass matrix $[M]$ in equation (4a) in terms of $\{q(t)\} = \{u_{ysaj}(t), u_{yca}(t), \theta_{xGa}(t), \theta_{zGa}(t), \theta_{xPa}(t), \theta_{zPa}(t), \theta_{zMa}(t), \theta_{zLa}(t), \theta_{xca}(t), \theta_{zca}(t)\}^T$ is

$$[M] = \text{diag} \{m_{sej}, m_{xc}, I_{xG}, I_{zG}, I_{xP}, I_{zP}, I_{zM}, I_{zL}, I_{xc}, I_{zc}\}. \quad (21)$$

The matrices $[K]_{11}$, $[K]_{12}$, $[K]_{13}$, $[K]_{22}$, $[K]_{23}$ and $[K]_{33}$ define the symmetric $[K]$ of dimension $2n_s + 11$ in equation (4b). Non-zero elements of $[K]_{11}$ of dimension $2n_s + 3$ are

$$k_{n_s+2, n_s+2} = k_{2n_s+2, 2n_s+2} = k_{11} = k_{n_s+1, n_s+1} = 12EI/L_e^3 + k_{byy}, \quad (22a)$$

$$k_{w_I+1, w_I} = k_{w_I, w_I+1} = -12EI/L_e^3,$$

$$k_{w_I+n_s+2, w_I+n_s+1} = k_{w_I+n_s+1, w_I+n_s+2} = -12EI/L_e^3, \quad w_I = 1, 2, 3, \dots, 2n_s, \quad (22b)$$

$$k_{w_{II}, w_{II}} = 24EI/L_e^3, \quad k_{w_{II}+n_s+1, w_{II}+n_s+1} = 24EI/L_e^3,$$

$$w_{II} = 2, 3, \dots, n_s/2, n_s/2+2, \dots, n_s, \quad (22c)$$

$$k_{n_s/2+1, n_s/2+1} = 24EI/L_e^3 + k_h, \quad k_{3n_s/2+2, 3n_s/2+2} = 24EI/L_e^3 + k_h,$$

$$k_{n_s/2+1, 3n_s/2+2} = k_{3n_s/2+2, n_s/2+1} = -k_h, \quad k_{2n_s+3, 2n_s+3} = 4k_{byy} + k_{yy}, \quad (22d)$$

$$k_{w_{III}, 2n_s+3} = k_{2n_s+3, w_{III}} = -k_{byy}, \quad w_{III} = 1, n_s+1, n_s+2, 2n_s+2. \quad (22e)$$

The non-zero elements of $[K]_{12}$ of $2n_s + 3$ rows and eight columns are

$$k_{3n_s/2-1, 1} = -k_{3n_s/2+3, 1} = k_{n_s/2, 3} = -k_{n_s/2+2, 3} = 6EI/L_e^2, \quad k_{2n_s+3, 7} = 4k_{by\theta_x},$$

$$-k_{n_s/2+1, 2} = k_{3n_s/2+2, 2} = k_h/(2d_G), \quad k_{n_s/2+1, 4} = -k_{3n_s/2+2, 4} = k_h/(2d_P), \quad (23a)$$

$$k_{w_{III}, 7} = -k_{by\theta_x} + z_{w_{III}} k_{byy}, \quad k_{w_{III}, 8} = -x_{w_{III}} k_{byy},$$

$$w_{III} = 1, n_s+1, n_s+2, 2n_s+2. \quad (23b)$$

The non-zero elements of $[K]_{13}$ of $2n_s + 3$ rows and $2n_s$ columns are

$$k_{11} = k_{n_s+2, n_s+1} = 6EI/L_e^2 + k_{by\theta_x}, \quad k_{n_s+1, n_s} = k_{2n_s+2, 2n_s} = -6EI/L_e^2 + k_{by\theta_x},$$

$$k_{3n_s/2+2, 3n_s/2+1} = -k_{3n_s/2+1, 3n_s/2} = k_{n_s/2+1, n_s/2+1} = -k_{n_s/2+1, n_s/2} = 6EI/L_e^2,$$

$$k_{2n_s+3, 1} = k_{2n_s+3, n_s} = k_{2n_s+3, n_s+1} = k_{2n_s+3, 2n_s} = -k_{by\theta_x}, \quad (24a)$$

$$\begin{aligned}
k_{12} &= -k_{n_s+1, n_s-1} = k_{n_s/2, n_s/2-1} = k_{n_s/2+2, n_s/2+2} = k_{n_s+2, n_s+2} = 6EI/L_e^2, \\
-k_{2n_s+2, 2n_s-1} &= -k_{3n_s/2+1, 3n_s/2-1} = k_{3n_s/2+3, 3n_s/2+2} = 6EI/L_e^2, \quad n_s > 2, \quad (24b) \\
-k_{w_{IV}, w_{IV}-1} &= k_{w_{IV}, w_{IV}+1} = k_{w_{IV}+n_s/2+1, w_{IV}n_s/2+1} = -k_{w_{IV}+n_s+1, w_{IV}-1+n_s} = 6EI/L_e^2, \\
k_{w_{IV}+n_s+1, w_{IV}+1+n_s} &= -k_{w_{IV}+n_s/2+1, w_{IV}+n_s/2-1} = k_{w_{IV}+3n_s/2+2, w_{IV}+3n_s/2+1} = 6EI/L_e^2, \\
-k_{w_{IV}+3n_s/2+2, w_{IV}+3n_s/2-1} &= 6EI/L_e^2, \quad n_s \geq 6, \quad w_{IV} = 2, 3, \dots, n_s/2-1. \quad (24c)
\end{aligned}$$

Non-zero elements given by equations (24b) and (24c) exist only if $n_s > 2$ and $n_s \geq 6$ respectively. The non-zero elements of symmetric $[K]_{22}$ of dimension eight are

$$\begin{aligned}
k_{11} &= k_{33} = 8EI/L_e, \quad k_{22} = k_{TM} + k_h d_G^2/4, \quad k_{25} = -k_{TM}, \quad k_{46} = -k_{TL}, \\
k_{44} &= k_{TL} + k_h d_P^2/4, \quad k_{24} = k_{42} = -k_h d_G d_P/4, \quad k_{55} = k_{TM}, \quad k_{66} = k_{TL}, \\
k_{77} &= 4k_{b\theta_x \theta_x} + \Sigma(z_j)^2 k_{b_{yy}} + k_{v\theta_x}, \quad k_{88} = k_{v\theta_z} + \Sigma(x_j)^2 k_{b_{yy}}, \quad (25)
\end{aligned}$$

where k_{TL} and k_{TM} are the effective torsional stiffnesses due to shaft and torsional coupling respectively at load and motor ends, and the summation Σ is over the four ball bearings denoted by $j = 1, n_s + 1, n_s + 2, 2n_s + 2$. The non-zero elements of $[K]_{23}$ of eight rows and $2n_s$ columns are

$$k_{1, n_s/2} = k_{1, n_s/2+1} = k_{3, 3n_s/2} = k_{3, 3n_s/2+1} = 2EI/L_e, \quad (26a)$$

$$k_{7, w_V} = +z_{w_{III}} k_{b_{y\theta_x}} - k_{b\theta_x \theta_x}, \quad k_{8, w_V} = -x_{w_{III}} k_{b_{y\theta_x}},$$

$$w_{III} = 1(w_V = 1), \quad n_s + 1(w_V = n_s), \quad n_s + 2(w_V = n_s + 1), \quad 2n_s + 2(w_V = 2n_s). \quad (26b)$$

Finally, the non-zero elements of symmetric $[K]_{33}$ of dimension $2n_s$ are

$$k_{11} = k_{n_s, n_s} = k_{n_s+1, n_s+1} = k_{2n_s, 2n_s} = 4EI/L_e + k_{b\theta_x \theta_x}, \quad (27a)$$

$$k_{w_{VI}, w_{VI}} = k_{n_s + w_{VI}, n_s + w_{VI}} = 8EI/L_e, \quad n_s > 2, \quad w_{VI} = 2, 3, \dots, n_s, \quad (27b)$$

$$k_{w_{VII}, w_{VII}-1} = k_{w_{VII}-1, w_{VII}} = k_{n_s + w_{VII}, n_s + w_{VII}-1} = 2EI/L_e,$$

$$k_{n_s + w_{VII}-1, n_s + w_{VII}} = 2EI/L_e, \quad n_s > 2, \quad w_{VII} = 2, 3, \dots, n_s \quad (w_{VII} \neq n_s/2 + 1). \quad (27c)$$

As discussed earlier, our proposed vibration model again differs from the conventional spur gear pair dynamic models essentially because of the $[K]_{bm}$ formulation. The simple model with casing and mount dynamics can be obtained from our proposed model by retaining only $k_{b_{xx}}$, $k_{b_{yy}}$ and $k_{b_{zz}}$ in $[K]_{bm}$.

6.3. EIGENSOLUTION

The natural frequencies ω_j and modes ϕ_j of the system of Figure 1 with parameters given in Tables 1 and 4 were computed by using the proposed theory, a dynamic finite element model, and the simple theory with and without casing mounts. The FEM model includes casing flexibility—this will be described later in example case II. Our proposed theory differs from FEM by less than $\pm 4\%$ for the first six ω_j and is within $\pm 10\%$ for ω_7 and ω_8 ,

TABLE 5
Eigensolution of example case II
 (a) *Comparison of natural frequencies ω_j in Hz*

Mode, ϕ_j	Proposed theory	FEM	Δ_1 (%)	Simple models		Δ_2 (%)
				Without casing	With casing	
1	0	0	0.0	0	0	0.0
2	31.0	31.0	0.0	29.8	29.1	6.1
3	480	480	0.0	Not predicted	480	0.0
4	526	525	-0.2	Not predicted	510	2.9
5	563	564	0.2	Not predicted	560	0.7
6	790	760	-3.9	666	650	14.5
7	1147	1093	-4.9	837	809	26.0
8	1208	1100	-9.8	694	681	38.1

Δ_1 (%) = $100 \times (\text{FEM} - \text{proposed theory}) / \text{FEM}$; Δ_2 (%) = $100 \times (\text{FEM} - \text{simple model}) / \text{FEM}$.

(b) *Summary of mode shapes*

Mode, ϕ_j	Description
1	Motor-gear-pinion-load system rigid-body torsional motion
2	Motor-gear-pinion-load system torsional motion
3	Casing rigid body rotational motion θ_{xca}
4	Casing rigid body translational motion u_{yca}
5	Casing rigid body torsional motion θ_{zca}
6	First shaft coupled transverse-torsional motion
7	Second shaft coupled transverse-torsional motion
8	Shaft transverse motion

as shown in Table 5(a). The corresponding natural modes are described in Table 5(b). In general, simple models predict lower natural frequencies and deviate substantially except for ω_1 and ω_3 . The simple model without any casing dynamics as outlined here is in excellent agreement with the results reported by Kahraman *et al.* [2]. This provides further validation for our theory.

The effect of casing mass m_c (and inertias I_{xc} , $I_{zc} \propto m_c$) is shown in Figure 4(a) for ω_3 , ω_5 and ω_7 . Here ω_2 and ω_6 are unaffected by m_c . On the other hand, ω_3 is always proportional to $(m_c)^{-1/2}$ except when $m_c / (m_s + m_G + m_P) \ll 1$. This proportionality feature, similar to that of the single-degree-of-freedom theory, is also seen for ω_7 and ω_8 for a lightweight casing, and ω_4 and ω_5 for a massive casing. It may be noted that the trends of ω_8 and ω_4 are similar to those of ω_7 and ω_5 respectively, although they are not included in Figure 4(a). Unlike example case I, in which the transition point is at $m_c / (m_s + m_G + m_P) = 1$, two transition points are observed here due to the effects of the gear, pinion and casing rotary inertias.

The effect of mount stiffness k_{vy} (and $k_{v\theta_x}$, $k_{v\theta_z} \propto k_{vy}$) on ω_j is given in Figure 4(b). One again finds that ω_2 and ω_6 are not affected by a variation in k_{vy} . Also, two transition points are found which separate a region of constant ω_j from the region where $\omega_j \propto (k_{vy})^{1/2}$ for some of the natural frequencies. Here ω_8 is similar to ω_7 , which is proportional to $(k_{vy})^{1/2}$ for stiff mounts but is insensitive to k_{vy} for softer mounts. The converse is seen for ω_2 with a lower transition point. On the other hand, $\omega_5 \propto (k_{vy})^{1/2}$ in the region between the two transition points, and is constant elsewhere. Finally, ω_3 is seen to be always increasing with k_{vy} .

The effect of mean bearing displacement δ_{zm} is summarized in Table 6 for two extreme δ_{zm} values. Only the natural frequencies associated with the shaft transverse and/or torsional motion are sensitive to δ_{zm} or preloads $F_{zbm}(\delta_{zm})$. The rate of increase for ω_j is high for low δ_{zm} , but the rate decreases with a higher δ_{zm} due to the stabilization of k_{bij} .

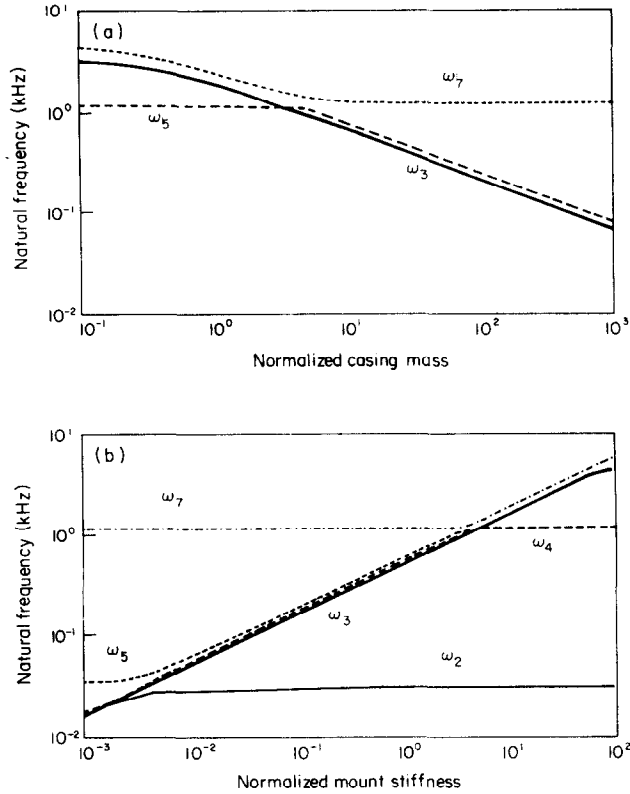


Figure 4. Effect of casing mass m_c (normalized with respect to shaft and gear masses) and mount stiffness k_{ux} (normalized with respect to $k_{b,xx}$) on the system natural frequencies ω_j for example case II.

TABLE 6

Example case II: effect of mean bearing displacement δ_{zm} on ω_j (Hz)

δ_{zbm} (mm)	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8
0.06	0	29.9	480	520	562	764	852	852
1.00	0	31.0	480	526	563	791	1180	1215

6.4. TRANSMISSIBILITY SPECTRA

The response due to the static transmission error excitation $e(t)$ at the mesh point is computed by using the same dynamic stiffness technique used for example case I. Here, only the fundamental harmonic of $e(t)$ at mesh frequency ω_h is included. All transmissibility functions are normalized with respect to the magnitude of the gear mesh force $F_h(t) = k_h e(t)$. The bearing force $R_{F_{yba}, F_h}(\omega_h)$ and moment $R_{M_{xba}, F_h}(\omega_h)$ transmissibility spectra, as predicted by our formulation and by the simple model, are compared in Figure 5(a) and (b). It may be noted that the transmissibility spectra for all four bearings are similar. Although the simple model results compare reasonably well with those of our proposed theory for $R_{F_{yba}, F_h}(\omega_h)$, it is not capable of predicting $R_{M_{xba}, F_h}(\omega_h)$, as also seen previously. In the case of the mount transmissibility shown in Figure 5(c), only the net moment $M_{zva}(t)$ is transmitted. The resultant vertical force $F_{yva}(t)$ and moment $M_{xva}(t)$ are negligible due to the force and moment cancellations at the mount feet. The Fourier spectrum of the normalized dynamic transmission error $p(t) = [y_P(t) - y_G(t) + (d_P \theta_P(t)/2) - (d_G \theta_G(t)/2)]/e(t)$ is shown in Figure 5(d). Only two modes, ϕ_2 which is predominantly

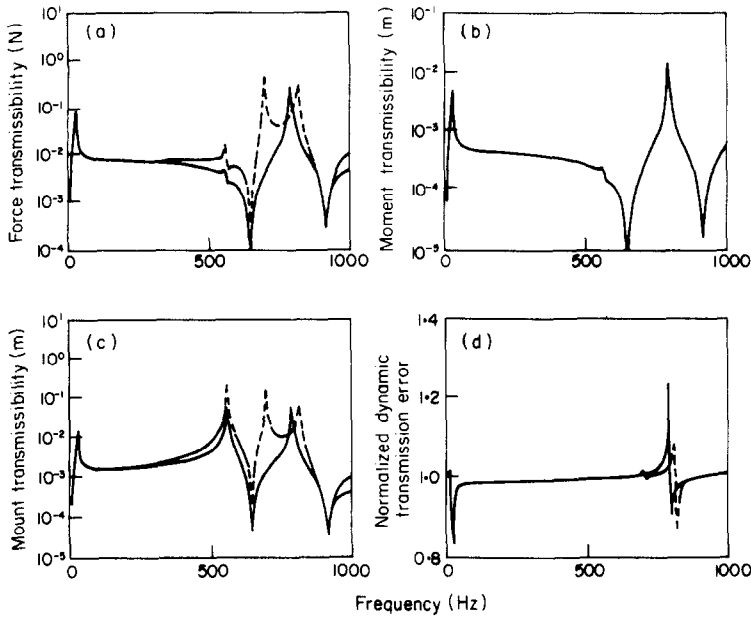


Figure 5. Comparison of transmissibility and normalized dynamic transmission error spectra for example case II. (a) Bearing force transmissibility $R_{F_{ba}, F_h}(\omega_h)$; (b) bearing moment transmissibility $R_{M_{ba}, F_h}(\omega_h)$ which is not predicted by simple model; (c) mount transmissibility $R_{M_{ba}, F_h}(\omega_h)$; (d) normalized dynamic transmission error $p(\omega_h)$. —, Proposed theory; ---, simple model.

torsional vibration of the shafts and ϕ_6 which is a coupled transverse-torsional mode, affect $p(\omega_h)$ significantly. The sixth mode produces a higher $p(\omega_h)$ amplitude at ω_h than ϕ_2 . Additionally, the simple theory underestimates $p(\omega_h)$ as compared to the proposed model.

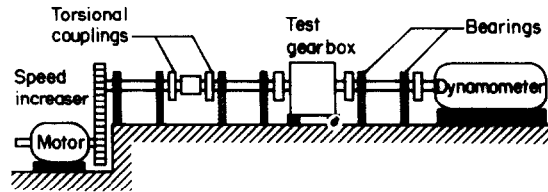
7. EXAMPLE CASE III: GEARED ROTOR SYSTEM WITH RIGIDLY MOUNTED FLEXIBLE CASING

7.1. PHYSICAL SET-UP

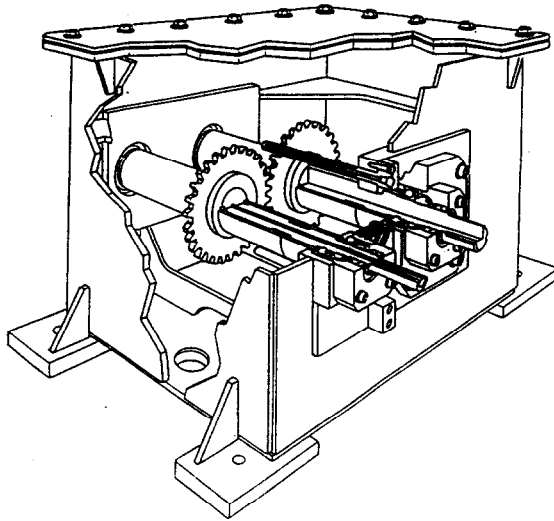
The final example case is that of the NASA gear test facility, as shown in Figure 6 [32]. The system parameters are equivalent to those in Tables 1 and 4 except for the flexible steel casing of approximate dimensions $0.33 \times 0.28 \times 0.25 \text{ m}^3$ and plate thickness of 0.006 m . Rigid mount feet attach the four corners of the bottom casing plate to a massive foundation. A high-precision gear and pinion are used, which are identical with 0.006 m facewidth, 0.089 m diameter and a 1:1 ratio. Four axially preloaded high-precision deep groove ball bearings are used to support 0.03 m diameter shafts of length 0.254 m on the flexible casing. The input and output shafts are only coupled torsionally to the rest of the gear test facility through flexible couplings, and the geared system is driven by a 149 kW d.c. motor. The vibrational level at various positions on the casing plate is measured by using PCB 303A and Endevco 2271 accelerometers over the operational speed range of $2250\text{--}5750 \text{ rpm}$, which corresponds to the gear mesh frequency ω_h range of $1050\text{--}2683 \text{ Hz}$.

7.2. VIBRATION MODELS

The dynamic finite element model for Figure 6(b) was developed by using commercial software [25], as outlined in section 4.1. The magnitudes of k_{bij} , which are included in the FEM model through a six-dimensional generalized stiffness matrix element, are computed



(a)



(b)

Figure 6. Example case III: (a) schematic of the NASA gear test facility; (b) NASA spur gear pair system supported on rigidly mounted flexible casing [32].

by using the two non-linear algebraic equations (19) and (20), in which the static elastic deformation of the casing plate is neglected. The shafts are modeled by using two-noded Timoshenko beam elements with axial degrees of freedom capability. Four-noded quadrilateral plate elements with shear deformation and rotary inertia effects are used to construct the flexible casing. The shafts and torsional couplings exterior of the test gearbox are modeled as torsional stiffness elements only, and the gear, pinion, motor and load are incorporated in the FEM model using generalized mass and inertia elements. The gear mesh coupling between the gear and pinion is described by a generic stiffness matrix of dimension six, similar to $[K]_{bm}$. A free rotational $\theta_{za}(t)$ boundary condition is specified at the motor and load inertias, while ideally clamped $u_{wa} = \theta_{wa} = 0$, $w = x, y, z$, boundary points at the corner of the bottom plate are assumed to simulate the rigid mount feet.

Natural frequencies and modes predicted by a similar FEM model have already been verified by comparison with experimental modal analysis in an earlier publication by us [16]. Hence, eigensolutions will not be discussed here. However, it may be noted that the conventional geared rotor dynamic models with simple bearing models and without casing dynamics are not valid for high excitation frequencies (beyond the first casing plate elastic mode). For the forced response study, only the fundamental harmonic ω_h of the static transmission error excitation $e(t) = 3.5 \sin(\omega_h t) \mu\text{m}$ computed from the gear tooth profile and geometry [33]. About 110 dynamic degrees of freedom are selected to minimize computational effort while still maintaining sufficient accuracy. A modal damping ratio of $\zeta = 0.05$ is assumed over the frequency range of interest.

7.3. CASING RESPONSE

Several locations on three different casing plate surfaces, as shown in Figure 7, have been chosen for the experimental validation of our theory. In Figure 8 predicted and

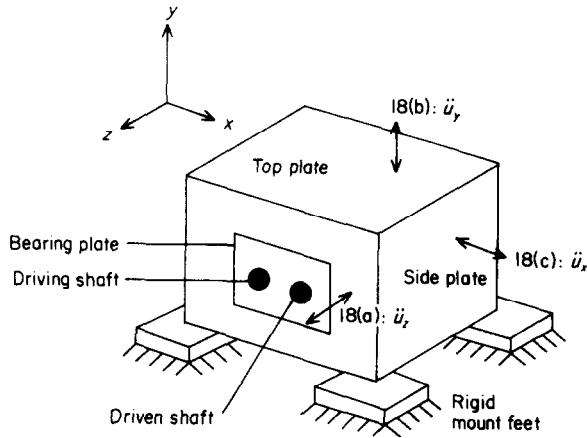


Figure 7. Measurement locations on the casing plate for example case III.

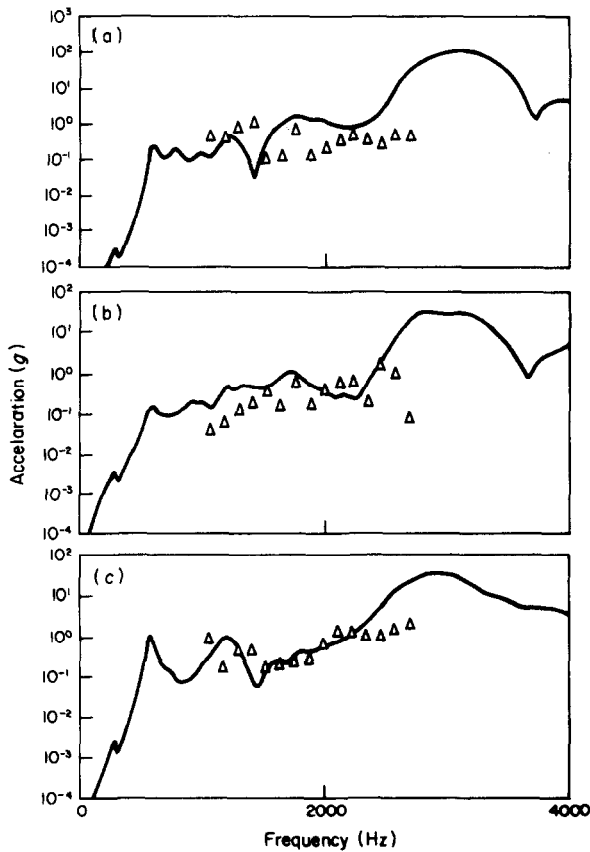


Figure 8. Comparison of casing flexural vibrations for example case III with $\delta_m = 0.08$ mm. (a) Side plate supporting the bearings and facing the dynamometer (location 18a); (b) top plate cover (location 18b); (c) side plate with no bearings (location 18c). Δ , Experiment; —, proposed theory.

measured mean square acceleration spectra at ω_b over 0–4 kHz are compared, even though measurements have been conducted only between 1 and 3 kHz. In general, good agreement is found between experiment and theory. Some discrepancy is observed above 2.4 kHz, due to the limitations of the FEM model in the high modal density regime. It may be noted that the simple theory in which only conventional bearing models are used cannot predict flexural vibration of the casing plate as shown in Figure 8. A broadband vibratory energy comparison is achieved by averaging the mean square value of the acceleration over one-third octave bands, L_A in dB per unit frequency bandwidth $\Delta\omega$, which is defined as

$$L_A \text{ (dB)} = 10 \log_{10} \left[\left(\frac{1}{2\Delta\omega} \right) \sum_{\Delta\omega} \operatorname{Re} \left\{ \frac{(AA^*)}{\langle A^2 \rangle_{ref}} \right\} \right], \quad \langle A^2 \rangle_{ref} = 1.0 \text{ g}^2, \quad (28)$$

where A^* denotes the complex conjugate of the acceleration. Our predictions are within ± 5.0 dB of the measured values, as shown in Table 7.

TABLE 7
Comparison of L_A (dB) for example case III

Center frequency (Hz)		Location in Figure 7		
		18(a)	18(b)	18(c)
1250	Theory	-3.0	-13	-4.5
	Experiment	-2.0	-17	-8.4
1600	Theory	-5.5	-7.0	-11
	Experiment	-10	-8.1	-14
2000	Theory	-11	-5.4	2.1
	Experiment	-10	-7.4	0.0
2500	Theory	-3.0	0.5	5.9
	Experiment	-7.3	-1.0	4.0

The sensitivity of results to $[K]_{bm}$ is evaluated next. Recall that $[K]_{bm}$ is computed on the basis of knowledge of the mean axial displacements, δ_{zm} , applied to the bearings. This parameter has been assumed to be a constant in our analysis, although its exact value for the NASA experimental set-up is not known. Using two reasonable values of δ_{zm} , we obtain the range of acceleration spectra shown in Figure 9. Almost all the experimental data are now within this range. This explains experimental scatter observed in Figure 8, as the experimental δ_{zm} may vary slightly from one steady state speed to another during testing.

In example case II, we found that only a few coefficients in $[K]_{bm}$ are necessary, depending on the excitation. Accordingly, we had used only k_{byy} , $k_{by\theta_x}$ and $k_{b\theta_x\theta_x}$. However, in the present FEM model of case III, we must include the entire symmetric $[K]_{bm}$ matrix of dimension six with no simplification at all. Now if we omit the off-diagonal term $k_{bx\theta_x}$, which is not directly excited by $e(t)$ in the y -direction of Figure 1, while still retaining other k_{bij} coefficients, no significant changes in our predictions are seen. Similarly, if the off-diagonal terms k_{bzy} and $k_{bz\theta_x}$, which are related to the shaft axial degrees of freedom, are also neglected, the predicted acceleration spectra vary slightly. However, if $k_{by\theta_x}$ and $k_{b\theta_x\theta_x}$, which constrain the bending motion of the shaft excited by $e(t)$, are assigned zero values as per the simple theory, zero out-of-plane acceleration levels are observed. This confirms that the vibration transfer through the bearings, from the shaft bending motion to the casing flexural motion, is highly dependent on these two terms.

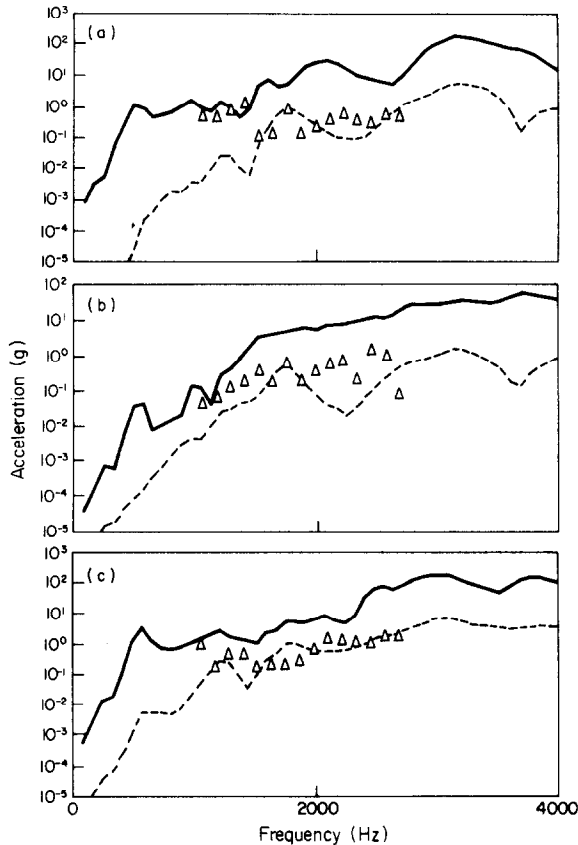


Figure 9. Sensitivity of predicted casing acceleration to mean bearing displacement for example case III. (a) Side plate supporting the bearings and facing the dynamometer (location (18a)); (b) top plate cover (location (18b)); (c) side plate with no bearings (location (18c)). Δ , Experiment; —, $2E-2$ mm; ---, $2E-1$ mm.

8. CONCLUDING REMARKS

A generic geared rotor system model has been developed, by using lumped parameter and dynamic finite element techniques, which incorporates the mathematical model for precision rolling element bearings proposed earlier in Part I. This system-oriented model includes the internal rotating system, the rolling element bearings, the flexible torsional couplings, the motor and load, a flexible or rigid casing, and compliant or massive mounts. The discrete shaft model includes torsional, flexural and longitudinal motions. In example cases I and II, only flexural and torsional motions are predicted. The dynamic analyses results for all three example cases indicate that our proposed model is clearly superior to the conventional simple models as given in the literature. For instance, we are able to predict bearing and mount moment transmissibilities and improve casing flexural vibration prediction significantly using our theory; measurements made on case III partially validate our formulations. Conversely, the simple models are unable to account for any bearing moment transmissibility, and consequently predict zero flexural response on the casing. The computational times for the lumped parameter models of the proposed and simple theories are nearly the same. However, the FEM is computationally more intensive, say three to five times longer, due to the large number of DOF in the flexible casing structure.

Our theory, although restricted to a linear and time-invariant dynamic system, is comprehensive. It can be used for analysis as well as design studies of other rotating mechanical

systems with multiple shafts and gear pairs or multi-staged rotors. We have also extended this formulation to examine the vibratory energy transfer over a broadband basis [34, 35].

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APPENDIX: LIST OF SYMBOLS

$\langle A^2 \rangle$	mean square acceleration spectra
$[C]$	system damping matrix
$[C]_b$	bearing damping matrix
DOF	degrees of freedom
d_G	gear diameter
d_P	pinion diameter
E_C	Rayleigh dissipation function
E_T	kinetic energy
E_U	potential energy
EI	flexural rigidity of the shaft
$e(t)$	static transmission error
$F_{jba}(t)$	alternating bearing force in the $j = x, y, z$ direction
F_{jbm}	mean bearing force in the $j = x, y, z$ direction
$F_{jsa}(t)$	applied alternating force on the shaft, $j = x, y, z$
$\bar{F}_w$	generalized force in Lagrangian formulation
$\{f(t)\}_a$	generalized alternating applied load vector
$\{f(t)\}_s$	total shaft load vector
$\{f(t)\}_{sa}$	alternating shaft load vector
G	shear modulus of elasticity
I_{jG}	mass moment of inertia of the gear about $j = x, y, z$ axis
I_{zL}	mass moment of inertia of the load about z axis
I_{zM}	mass moment of inertia of the motor about z axis

I_{jp}	mass moment of inertia of the pinion about $j=x, y, z$ axis
I_{jc}	mass moment of inertia of the casing about $j=x, y, z$ axis
$\{I\}$	lumped inertia row vector
K_n	rolling element load-deflection stiffness constant
$[K]$	system stiffness matrix
$[K]_{bm}$	proposed bearing stiffness matrix of dimension six
$[K]_h$	gear mesh coupling stiffness matrix
$[K]_s$	shaft stiffness matrix
$[K]_s^e$	shaft segment stiffness matrix
k_{TM}	effective torsional stiffness of the driving shaft coupling
k_{TL}	effective torsional stiffness of the driven shaft coupling
k_{vj}	mount stiffness coefficient, $j=x, y, z, \theta_x, \theta_y, \theta_z$
k_{bwj}	bearing stiffness coefficient, $w, j=x, y, z, \theta_x, \theta_y, \theta_z$
k_h	gear mesh stiffness
L_A	frequency bandwidth-averaged mean square acceleration in dB $re\langle A^2 \rangle_{ref} = 1.0 \text{ g}^2$
L_s	shaft segment length
$[M]$	system mass matrix
M_{jba}	alternating bearing moment about $j=x$ or y direction
M_{wbm}	mean bearing moment about $w=x$ or y direction
m_R	rotor mass
m_s	shaft mass
m_{sej}	shaft lumped mass
m_c	casing mass
$\{m\}$	lumped mass row vector
n	rolling element load-deflection exponent
n_s	number of shaft segments
$p(t)$	normalized dynamic transmission error
$\{q(t)\}_a$	generalized alternating displacement vector
$\{q\}_{bm}$	mean bearing displacement vector
$\{q(t)\}_{ca}$	alternating casing displacement vector
$\{q(t)\}_{sa}$	alternating shaft displacement vector
$\{q\}_{sm}$	mean shaft displacement vector
q_w	generalized displacement vector in Lagrangian formulation
Re	real part of a complex number
$\mathbf{R}_j$	j th bearing co-ordinate vector, $j=1, n_s+1, n_s+2, 2n_s+2$
r_L	bearing radial clearance
T_{ism}	applied mean torque on the shaft, $j=x, y$ or z .
t	time
$u_{jca}(t)$	alternating casing translational displacement in the $j=x, y$ or z direction
$u_{jsa}(t)$	alternating shaft translational displacement in the $j=x, y$ or z direction
$\{u(t)\}_{ca}$	casing alternative displacement vector
$\{u(t)\}_{sa}$	shaft alternating displacement vector
Z	total number of rolling element
α_0	unloaded bearing contact angle
β_{wa}	alternating bearing angular displacement about the $w=x$ or y direction
β_{wm}	mean bearing angular displacement about the $w=x$ or y direction
$\Delta\omega$	frequency bandwidth
δ_{ja}	alternating bearing translational displacement in the $j=x, y$ or z direction
δ_{jm}	mean bearing translational displacement in the $j=x, y$ or z direction
ϕ_j	mode shape vector corresponding to the j th natural frequency
$\theta_{jca}(t)$	alternating casing angular displacement about the $j=x, y$ or z co-ordinate
$\theta_{jsa}(t)$	alternating shaft angular displacement about the $j=x, y$ or z co-ordinate
$\{\theta(t)\}_{ca}$	casing alternating angular displacement vector
$\{\theta(t)\}_{sa}$	shaft alternating angular displacement vector
$\{\theta(t)\}_{Ra}$	rotor alternating angular displacement vector
σ	Rayleigh damping matrix proportionality constant
Ω_c	mean rotational speed of the shaft
Ω_{cM}	mean rotational speed of the driving shaft
Ω_{cL}	mean rotational speed of the driven shaft
ω_h	gear mesh frequency

ω_j	undamped natural frequency, $j = 1, 2, 3, \dots$
ζ_j	modal damping ratio
$[]^T$	transpose of a matrix or vector
$(\dot{})$	first time derivative
$(\ddot{})$	second time derivative
$()^*$	complex conjugate